

Deterministic pose-spline dynamics estimator

Shared parameter update: nominal -> estimated

mass [kg]	2.351597591 -> 2.778856645	delta=+0.427259
ratio=1.1816888		
CoG x [m]	0 -> -0.004276539835	delta=-0.00427654 ratio=-
CoG y [m]	0 -> 0.001447587676	delta=+0.00144759 ratio=-
CoG z [m]	0 -> -0.01354005303	delta=-0.0135401 ratio=-
rotor effectiveness 1	1 -> 1.008259485	delta=+0.00825949 ratio=1.0082595
rotor effectiveness 2	1 -> 1.003215156	delta=+0.00321516 ratio=1.0032152
rotor effectiveness 3	1 -> 1.000306584	delta=+0.000306584 ratio=1.0003066
rotor effectiveness 4	1 -> 0.9883265717	delta=-0.0116734 ratio=0.98832657
command lag [s]	0.01 -> 0.04764256869	delta=+0.0376426 ratio=-

Inertia matrix [kg m²]

nominal:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06633521267, 8.872587599e-05, 3.299651266e-05]
[ 8.872587599e-05, 0.1117651505, 2.083815787e-05]
[ 3.299651266e-05, 2.083815787e-05, 0.1771373101]
```

Principal moments: nominal -> estimated

principal moment 1 [kg m ²]	0.06495264516 -> 0.06633502959	delta=+0.00138238
ratio=1.021283		
principal moment 2 [kg m ²]	0.06500006701 -> 0.1117653171	delta=+0.0467653
ratio=1.7194646		
principal moment 3 [kg m ²]	0.1289921163 -> 0.1771373265	delta=+0.0481452
ratio=1.3732415		

Optimization

joint dynamics loss: 0.0923068235683
soft-prior cost: 0.0244477234859
selected strict-ZOH lag [s]: 0.047642568686

Bag failure_2_single

reconstruction position RMSE [m]: 0.000368116986919
reconstruction position component RMSE [m]: [0.000186 0.000244 0.000204]
reconstruction orientation RMSE [deg]: 0.245016993583
gyro consistency RMSE xyz [rad/s]: [0.028083 0.036566 0.033602]
specific-force consistency RMSE xyz [m/s²]: [0.315599 0.09794 0.08401]
inferred external force RMS xyz [N]: [0.204973 0.383891 0.67793]
inferred external torque RMS xyz [N m]: [0.038651 0.198979 0.065283]
bag dynamics loss: 0.0923068235683